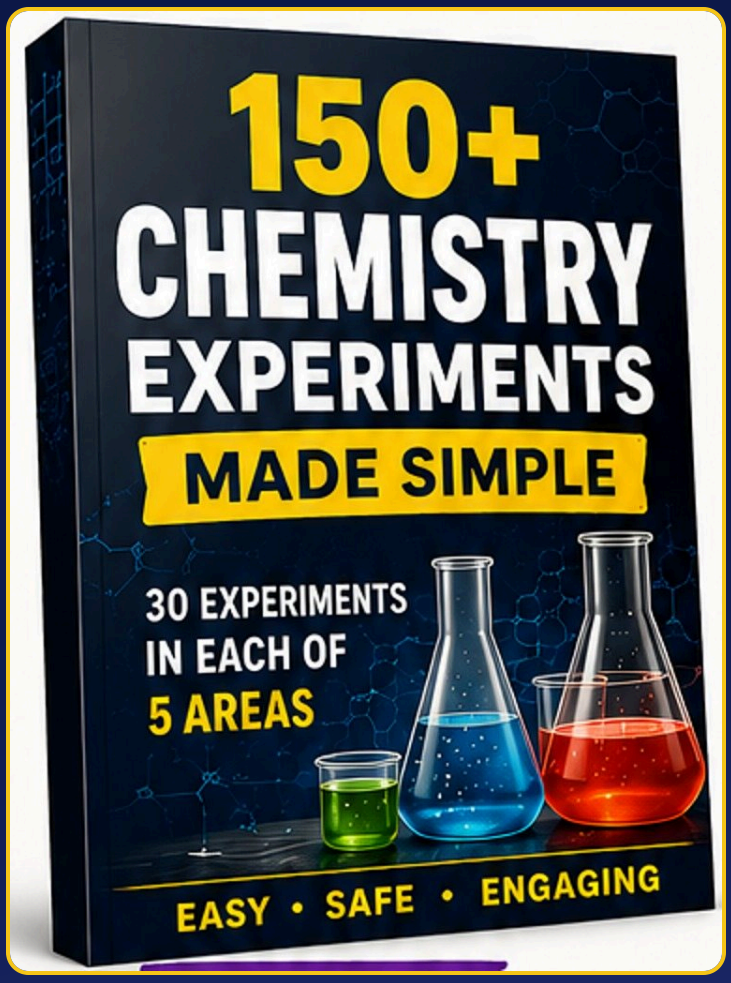
BONUS 1

Science Fair Kit - Plan, Present & Win

4-week timeline, 10 wow projects, board layout, header signs and judging rubric



In this guide

Welcome to the Science Fair Kit

How a Science Fair Works and What Judges Look For

The Scientific Method in Plain English

Choosing a Question

The 4-Week Timeline

Ten Wow Project Plans

Data and Graphs

The Research Notebook

Board Layout

Header Signs With Sample Text

Presentation Scripts

Common Judge Questions and Sample Answers

The Judging Rubric

The Final Checklist

Next Step

Board layout diagram

Welcome to the Science Fair Kit

> *A science fair is the one school event where a question you actually care about turns into a project you can hold, measure and explain. This kit walks you from a blank page to a finished board in four weeks.*

This kit is written for three readers at once. **Students** will find a step-by-step path: pick a question, test it fairly, record what happens and explain it with confidence. **Teachers** will find a ready-made timeline, a judging rubric and a set of ten project plans that need nothing more than household supplies. **Parents** will find clear guidance on what to supervise, what to leave to the student and what safety rules never change.

Every project in this kit uses the same style as the experiment cards in the main pack: a plain-words question, a hypothesis you can test, exact amounts, and a safety note. Nothing here needs a laboratory. Everything is built from items found in a grocery store, pharmacy or hardware store.

> **How to use this kit**
>
> Read the first six sections once, from start to finish. Then choose one of the ten project plans (or use the question checklist to invent your own) and follow the 4-week timeline day by day. Come back to the board, script and rubric sections in week 4.

A note for teachers

Print the timeline and the rubric for every student. Collect the parent permission form before any hands-on work begins. Use the checklist at the end of the kit as a sign-off sheet: a project is ready for the fair only when every box is ticked.

A note for parents

Your job is to supervise, ask questions and say yes to a safe plan. It is not your job to do the experiment, build the board or write the conclusion. Judges can tell, and the student learns far more from a messy honest result than from a perfect borrowed one. Stay in the room for every experiment, keep younger siblings and pets away from the work area, and make sure nothing that was used in a project is eaten or tasted.

Safety rules that apply to every project

- **Splash goggles on** for every step, from first pour to final cleanup.
- **An adult is present** for the whole experiment. Adults handle hot water, scissors and any candle.
- **Parent or guardian permission** is signed before the project starts.
- **Nothing from an experiment is ever eaten, drunk or tasted.** Use containers that will never be used for food again, or wash them thoroughly afterward.
- **Work in a ventilated area** on a tray or a covered table, and keep paper towels handy.
- **Only the materials listed.** Never add other cleaners, acids, bases or chemicals. Never mix household cleaners.
- **Wash hands** with soap and water when finished.

How a Science Fair Works and What Judges Look For

Most school fairs follow the same pattern. Students pick a question, test it over a few weeks and present the result on a standing display board next to their notebook and, sometimes, a small demonstration. On fair day, one or two judges visit each project, listen to a short talk and ask questions. Then they score the project on a rubric and give feedback.

The day of the fair, step by step

1. **Setup (30-60 minutes).** Students arrive, place the board on their table and arrange the notebook, a sample of materials and any data tables.
2. **Judging (1-3 hours).** Judges walk the room. Each visit lasts about five minutes: a short talk from the student, then questions.
3. **Public viewing.** Families and classmates walk through and ask their own questions.
4. **Awards and cleanup.** Results are announced, feedback sheets are returned and projects are taken down.

What judges actually look for

Judges are not looking for the biggest volcano or the most expensive materials. They are looking for evidence of **scientific thinking**. In plain words, they want to see that you asked a clear question, tested it fairly, collected honest data and drew a conclusion that matches what you found.

WHAT JUDGES LOOK FOR	WHAT IT LOOKS LIKE ON YOUR PROJECT
A clear, testable question	One sentence that can be answered by measuring something
A fair test	Only one thing changed on purpose; everything else kept the same
Real data	Numbers in a table, repeated trials, honest measurements
Understanding	You can explain WHY it happened, not only WHAT happened
Honesty	You report the result you got, even if it did not match your guess
Communication	A tidy board, clear graphs and a confident, calm talk
Ownership	You did the work yourself and can answer questions about it

> **The most important secret**
>
> A project whose hypothesis turned out wrong, but which was tested carefully and explained honestly, will almost always score higher than a perfect-looking project with no real data.

The Scientific Method in Plain English

The scientific method is not a rigid recipe. It is a habit: ask, guess, test, look, think, share. Here are the steps in everyday words.

STEP	IN PLAIN ENGLISH	EXAMPLE (BAKING SODA AND VINEGAR)
1. Ask a question	What do I want to find out?	Does more vinegar make a bigger fizz?
2. Do background research	What do I already know, and what have others found?	Vinegar and baking soda make carbon dioxide gas.
3. Make a hypothesis	My best guess, with a reason, in an if-then-because sentence	If I use more vinegar, then the balloon will inflate more, because there is more acid to react.
4. Test with an experiment	A fair test that changes one thing and keeps the rest the same	Same baking soda each time; change the vinegar from 1/4 cup to 1 cup.
5. Collect data	Measure and write down what happens	Measure the balloon width in inches each time.
6. Analyze the data	Make tables and graphs; look for patterns	Average three trials and graph them.
7. Draw a conclusion	Was my hypothesis supported? Why or why not?	More vinegar did help, but only up to a point, because the baking soda ran out.
8. Share the results	Show others and answer questions	Present at the fair.

What is a hypothesis, really?

A hypothesis is a prediction you can test, with a reason attached. A good format is: *If I change [the one thing], then [this measurement] will [go up, go down or stay the same], because [the science idea].* A hypothesis is not a question and it is not a fact. It is allowed to be wrong. Being wrong teaches you something new.

Variables: the three kinds you must know

- **Independent variable:** the one thing you change on purpose.
- **Dependent variable:** the thing you measure to see the effect.
- **Controlled variables:** everything you keep the same so the test stays fair.

If you change two things at once, you will never know which one caused the result. That is the single most common mistake at science fairs, and the easiest to fix.

> **Fair test check**
>
> Before you start, ask: 'If my result changes, can I point to exactly one reason?' If the answer is no, find the extra thing that changed and control it.

Choosing a Question

The best project question is one you are curious about, that you can answer by measuring, with materials that are safe and easy to get. Start with something you noticed in daily life: why do apples turn brown, why does the pool smell different, why does ice melt faster on a metal plate?

IS THIS A GOOD QUESTION? TICK EVERY BOX

☐ I can say it in one sentence.

☐ It starts with words like 'How does', 'Does', 'Which' or 'What happens to'.

☐ I can answer it by measuring something with a number (time, height, temperature, color scale, mass).

☐ I can change just one thing on purpose.

☐ I can repeat the test at least three times.

☐ It only needs safe household materials and an adult can supervise.

☐ It is not answered by a quick internet search (it needs my own data).

☐ I am actually curious about the answer.

Turn a topic into a testable question

TOPIC	TOO VAGUE	TESTABLE QUESTION
Rust	How does rust work?	Which household liquid makes an iron nail rust fastest?
Crystals	Crystals are cool	Does water temperature change how fast sugar crystals grow?
Fizz	Baking soda is fun	How does the amount of yeast change how much foam a reaction makes?
Water	Is water clean?	How well do different filter layers remove dirt from muddy water?
Food	Vitamin C is healthy	Which juice contains the most vitamin C?
Colors	Cabbage turns colors	What pH range do ten household liquids show on a cabbage indicator?

Questions to avoid

- Questions that need strong chemicals, open flames or special equipment.
- Questions about tasting, eating or testing on people or animals.
- Questions with only a yes or no answer and nothing to measure.
- Questions where you cannot control the variables (for example 'Which cloud shape means rain?').

If you do not want to invent your own, use one of the ten project plans in this kit. Each one is a ready-to-run question with materials, steps and a data table. You can also modify any plan: swap the liquids, change the temperatures, or compare a different set of materials. A small change makes it your own.

The 4-Week Timeline

Four weeks is enough time for a strong project if you work a little each day. The plan below assumes about 20-40 minutes a day on weekdays and a longer session once a week. If your fair is sooner, combine days; if it is later, repeat trials.

Week 1: Choose and Research

DAY	DAILY TASK
Monday	Brainstorm 5 things you are curious about. Write them in your notebook.
Tuesday	Use the question checklist to pick the best one. Write it as a testable question.
Wednesday	Research: read two sources about the science behind it. Write 5 key facts in your own words.
Thursday	Write your hypothesis (if-then-because). List the independent, dependent and controlled variables.
Friday	Make a materials list. Show your plan to an adult and ask for parent permission.
Weekend	Shop for materials. Set up a clean work area and start your notebook with the title page.

Week 2: Plan and Test

DAY	DAILY TASK
Monday	Write the procedure as numbered steps. Make the blank data table in your notebook.
Tuesday	Do a small practice run with an adult. Note anything that was confusing or unfair.
Wednesday	Fix the plan: change amounts, timing or tools. Rewrite the procedure.
Thursday	Check safety: goggles, tray, ventilation, adult present, cleanup plan.
Friday	Do a second practice run. Take clear photos of the setup (no faces needed).
Weekend	Finalize the procedure and make a checklist for experiment days.

Week 3: Experiment and Record

DAY	DAILY TASK
Monday	Run trial 1 for every setting. Record every number immediately.
Tuesday	Run trial 2. Keep the same tools, amounts and timing.
Wednesday	Run trial 3. Take photos at the same distance and angle each time.
Thursday	Calculate averages. Look for patterns and for results that do not fit.
Friday	Repeat any trial that went wrong. Make rough graphs by hand.
Weekend	Write your conclusion: was your hypothesis supported? Why? What would you do next?

Week 4: Board and Rehearse

DAY	DAILY TASK
Monday	Make final graphs. Print or write the header signs.

DAY	DAILY TASK
Tuesday	Print photos and glue everything onto the board panels.
Wednesday	Write your 2-minute talk. Practice it out loud once.
Thursday	Practice the 5-minute talk and answer five judge questions with a family member.
Friday	Pack the notebook, board, samples and a small repair kit (tape, glue stick, scissors, markers).
Fair day	Arrive early, set up, take a breath and smile. You know this project better than anyone.

> **If you fall behind**
>
> Protect the experiment days in week 3 first. A project with good data and a simple board beats a beautiful board with thin data.

Ten Wow Project Plans

Each plan below follows the same layout: question, hypothesis example, materials with amounts, variables table, procedure, data table to fill, what to graph, expected result in plain words, a safety note and an extension. Use goggles, supervision and permission for every plan. Use 3% hydrogen peroxide from a pharmacy at most. No plan uses flames, strong acids or strong bases.

Project 1: Elephant Toothpaste Catalysts

Question: Which catalyst makes the most foam from 3% hydrogen peroxide: dry yeast, a chopped raw potato or a chopped raw carrot?

Hypothesis example: If I use dry yeast, then the foam will be tallest, because yeast contains a lot of catalase, an enzyme that speeds up the breakdown of hydrogen peroxide.

Materials (all household and safe):

- 1/2 cup (120 mL) 3% hydrogen peroxide per trial (pharmacy strength, no stronger)
- 2 tsp (10 mL) warm water
- 1 tsp (4 g) dry yeast
- 1/4 cup (about 40 g) finely chopped raw potato
- 1/4 cup (about 30 g) finely grated raw carrot
- 1 tbsp (15 mL) dish soap
- A clean 16 oz (500 mL) plastic bottle and a tray with a rim
- A ruler and a timer

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Type of catalyst (yeast, potato, carrot)

VARIABLE TYPE	IN THIS PROJECT
Dependent (what you measure)	Foam height in inches (cm) after 2 minutes
Controlled (keep the same)	Hydrogen peroxide amount, soap amount, bottle size, starting temperature, catalyst amount, timing

Procedure:

1. Put on goggles. Stand the bottle in the middle of the tray.
2. Pour 1/2 cup (120 mL) of 3% hydrogen peroxide into the bottle and add 1 tbsp (15 mL) of dish soap. Swirl gently.
3. Mix the yeast with 2 tsp (10 mL) of warm water in a small cup and let it sit for 2 minutes. An adult handles the warm water.
4. Pour the catalyst into the bottle and step back. Start the timer.
5. At 2 minutes, measure the foam height against the ruler placed beside the bottle. Record it.
6. Rinse everything with plenty of water, dry the bottle, and repeat with the next catalyst. Do 3 trials for each.

Data table to fill in:

CATALYST	TRIAL 1 HEIGHT	TRIAL 2 HEIGHT	TRIAL 3 HEIGHT	AVERAGE
Dry yeast				
Raw potato				
Raw carrot				

What to graph: A bar graph with catalyst on the bottom axis and average foam height on the side axis.

Expected result in plain words: Yeast usually makes a tall, fast foam. Potato and carrot make a slower, smaller foam, because they hold fewer of the helper molecules that break hydrogen peroxide into water and oxygen gas. The soap traps the oxygen as bubbles.

> **Safety note**
> Use only 3% hydrogen peroxide. Keep it away from eyes, and wipe up spills right away. Wear gloves if you have them and wash hands after touching the foam. Never taste anything. Do not seal the bottle.

Extension: Repeat with the yeast at different water temperatures to see whether warmer yeast works faster. Keep water warm, never hot.

Project 2: Red Cabbage pH of Household Liquids

Question: What pH range does red cabbage juice show for ten common household liquids?

Hypothesis example: If a liquid is more acidic, then the cabbage juice will turn pinker or redder, because the pigment in cabbage changes its shape and color with the amount of hydrogen ions.

Materials (all household and safe):

- 1/4 head of red cabbage, chopped
- 2 cups (480 mL) hot water, handled by an adult
- A strainer and a clean jar
- 10 small clear cups
- 1 tbsp (15 mL) each of: white vinegar, lemon juice, clear soda, milk, tap water, baking soda solution (1 tsp in 1/2 cup water), dish soap solution (1 tsp in 1/2 cup water), washing soda solution (1/2 tsp in 1/2 cup water), clear shampoo solution (1 tsp in 1/2 cup water), plain seltzer
- Labels, a marker and a white sheet of paper for a background

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	The household liquid being tested
Dependent (what you measure)	The color of the cabbage indicator (rated on a color scale you make)
Controlled (keep the same)	Amount of indicator (2 tbsp, 30 mL, per cup), amount of liquid, lighting, background, cup size

Procedure:

1. Put on goggles. An adult pours hot water over the chopped cabbage in a heat-safe bowl and lets it sit for 15 minutes.
2. Strain the purple liquid into a jar. This is your indicator. Let it cool.
3. Label the 10 cups and set them on the white paper.
4. Pour 2 tbsp (30 mL) of indicator into each cup.
5. Add 1 tbsp (15 mL) of each liquid to its cup. Stir with a clean spoon each time.
6. After 1 minute, photograph the row and draw each color in your notebook. Rank them from most acidic (pink) to most basic (green or yellow).

Data table to fill in:

LIQUID	COLOR SEEN	RANK (1 = MOST ACIDIC)	ACID, NEUTRAL OR BASE?
White vinegar			
Lemon juice			
Tap water			
Baking soda solution			

What to graph: A color strip chart: draw the ten colors in order from red to green, labeled with the liquid names.

Expected result in plain words: Acids such as vinegar and lemon juice turn the purple juice pink or red. Neutral liquids stay purple. Baking soda and soap solutions turn it blue to green. The colors are a clue, not an exact pH number.

> **Safety note**
> Do not mix the test liquids together and do not use cleaning products like bleach or drain cleaner. Do not taste anything. Wipe up cabbage juice because it stains.

Extension: Make your own paper test strips by soaking coffee filter strips in the indicator and letting them dry.

Project 3: Crystal Growth vs Temperature

Question: Does the temperature of the water change how big sugar crystals grow in 7 days?

Hypothesis example: If I make the solution with warmer water, then bigger crystals will grow, because more sugar dissolves in warmer water, so more is left to form crystals as it cools.

Materials (all household and safe):

- 3 clean glass jars
- 3 cups (about 600 g) sugar
- 1 cup (240 mL) water per jar
- 3 wooden skewers and 3 clothespins
- A thermometer
- A saucepan and stove, used only by an adult
- Masking tape and a marker
- A ruler

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Temperature of the water when the sugar is dissolved (140 F / 60 C, 170 F / 77 C, 200 F / 93 C)
Dependent (what you measure)	Length of the longest crystal in millimeters after 7 days
Controlled (keep the same)	Amount of sugar and water, jar size, skewer type, room, light, how long the jars sit

Procedure:

1. Put on goggles. An adult heats water to the target temperature and measures it with the thermometer.
2. Stir in sugar, 1/2 cup (100 g) at a time, until no more dissolves. Use the same total amount in all three jars.
3. Pour into the jars. Wet a skewer, roll it in dry sugar and let it dry.
4. Hang the skewer from a clothespin so it dips into the liquid without touching the bottom.
5. Label the jars, cover loosely with paper towel and leave undisturbed for 7 days.
6. Measure the longest crystal on each skewer and record the length each day if possible.

Data table to fill in:

JAR TEMPERATURE	DAY 3 LENGTH	DAY 5 LENGTH	DAY 7 LENGTH	NOTES
140 F / 60 C				
170 F / 77 C				
200 F / 93 C				

What to graph: A line graph with days on the bottom axis and crystal length on the side axis, one line for each temperature.

Expected result in plain words: Warmer solutions usually hold more sugar, so more crystal grows as the water slowly cools and evaporates. Your result may differ because dust, vibration and cooling speed also matter, and you can discuss this as sources of error.

> **Safety note**
>
> Hot sugar syrup causes severe burns. An adult does all heating and pouring. Do not taste the crystals. Keep jars out of reach of small children and pets.

Extension: Try salt or Epsom salt in place of sugar and compare the crystal shapes with a magnifying glass.

Project 4: Rate of Reaction vs Surface Area

Question: Does breaking an effervescent tablet into smaller pieces make it dissolve faster?

Hypothesis example: If I crush the tablet into powder, then it will dissolve fastest, because more of its surface touches the water at once.

Materials (all household and safe):

- 4 effervescent antacid tablets (same kind), from a pharmacy
- 4 clear cups
- 1 cup (240 mL) of room-temperature water per trial
- A zip bag and a spoon for crushing
- A timer
- A thermometer to check water temperature

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Form of the tablet (whole, halves, quarters, powder)
Dependent (what you measure)	Time to fully dissolve, in seconds
Controlled (keep the same)	Water volume, water temperature, tablet brand, cup size, stirring (none)

Procedure:

1. Put on goggles. Check that all water is the same temperature.
2. Prepare four forms: one whole tablet, one in halves, one in quarters, one crushed to powder in the bag.

3. Pour 1 cup (240 mL) of water into each cup.
4. Drop the first form in and start the timer. Do not stir.
5. Stop the timer when the fizzing stops and the water is clear. Record the time.
6. Repeat with new tablets for 3 trials of each form.

Data table to fill in:

TABLET FORM	TRIAL 1 (S)	TRIAL 2 (S)	TRIAL 3 (S)	AVERAGE (S)
Whole				
Halves				
Quarters				
Powder				

What to graph: A bar graph of the average time for each form. Optionally add a line graph with the number of pieces on the bottom axis.

Expected result in plain words: The more pieces, the faster the dissolving, because tiny pieces expose more surface to the water so more particles collide every second. The whole tablet is the slowest.

> **Safety note**
> Do not swallow any of the tablet or the water. Use clean cups that will not be used for drinks again. Dispose of the solution down the drain with plenty of water.

Extension: Repeat the test with warm, room-temperature and cold water to compare the effect of temperature.

Project 5: Water Filter Effectiveness

Question: Which filter layers make muddy water the clearest?

Hypothesis example: If a filter has sand, gravel and cotton layers, then the water will be clearest, because each layer traps different sized particles.

Materials (all household and safe):

- 3 clear plastic bottles (16 oz / 500 mL), cut in half by an adult
- 2 cups (about 400 g) clean sand
- 1 cup (about 200 g) small gravel
- 1 cup (about 50 g) cotton balls
- 3 coffee filters
- 3 cups (720 mL) muddy water: 3 cups water mixed with 3 tbsp soil
- Rubber bands, clear cups, and a flashlight
- A ruler and a white card for comparing clarity

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Type of filter (cotton only, sand only, sand + gravel + cotton + coffee filter)
Dependent (what you measure)	Water clarity, rated 1 to 5 against a card, or the depth at which a black line can be seen through the water
Controlled (keep the same)	Starting muddy water amount and mixture, pouring speed, bottle size, lighting, cup type

Procedure:

1. Put on goggles. An adult cuts the bottles and the cut ends become funnels placed upside down in clear cups.
2. Build three filters: cotton only, sand only, and layers (from bottom to top: coffee filter, cotton, sand, gravel).
3. Mix the muddy water well and measure 1 cup (240 mL) for each filter.
4. Pour slowly and evenly. Start the timer.
5. Record how long the water takes to drip through and compare clarity under the flashlight.
6. Repeat for three trials.

Data table to fill in:

FILTER TYPE	TIME (S)	CLARITY SCORE (1-5)	NOTES
Cotton only			
Sand only			
Layered			

What to graph: A bar graph of clarity by filter type, and a second one for time.

Expected result in plain words: The layered filter usually gives the clearest water, but it is also slower. A trade-off between speed and clarity is a great topic for your conclusion. Filtered water is NOT safe to drink.

> **Safety note**
> Never drink the filtered water. Filters like this remove dirt but not germs or chemicals. Use a tray, and have an adult cut the bottles.

Extension: Add a layer of crushed charcoal from a pet-store aquarium filter and test whether it removes color from tea.

Project 6: Rusting Race

Question: Which household liquid makes an iron nail rust fastest?

Hypothesis example: If a nail sits in salt water, then it will rust fastest, because dissolved salt helps electrons move and speeds up rusting.

Materials (all household and safe):

- 5 plain iron or steel nails (not galvanized), same size
- 5 clear cups
- 1/2 cup (120 mL) each of: tap water, salt water (1 tsp in 1/2 cup), white vinegar, cooking oil, dry (no liquid)
- Labels and a marker
- A camera or phone, and a ruler
- A tray

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Liquid around the nail
Dependent (what you measure)	Amount of rust, scored 0 to 5, and photos each day
Controlled (keep the same)	Nail type and size, cup size, liquid amount, location, light, number of days

Procedure:

1. Put on goggles. Clean each nail with a paper towel to remove oils.
2. Label the cups and add the liquids. Leave one cup with a nail and nothing else.
3. Place one nail in each cup so it is half covered.
4. Photograph all cups on day 0, with the same background and distance.
5. Check and score once a day for 7 days.
6. Compare the photos and the rust scores.

Data table to fill in:

LIQUID	DAY 1	DAY 3	DAY 5	DAY 7
Tap water				
Salt water				
Vinegar				
Cooking oil				
Dry				

What to graph: A line graph of the rust score for each day, one line per liquid.

Expected result in plain words: Salt water and vinegar usually rust the nail fastest. Oil protects it by keeping water and air away. The dry nail changes least. Rust is iron combining with oxygen and water.

> **Safety note**
> Rusty nails and the old liquid should be thrown away by an adult with gloves. Keep nails away from small children. Never touch the rust with an open cut.

Extension: Wrap a nail in a strip of a different metal, like aluminum foil, and see whether it changes the rust.

Project 7: Density Columns

Question: Does the density of a liquid decide where it sits in a stacked column?

Hypothesis example: If a liquid has more sugar dissolved in it, then it will sink below the less sugary ones, because the same space holds more mass.

Materials (all household and safe):

- 1 tall clear glass or jar
- 1/2 cup (120 mL) each of: honey, dish soap, water with food coloring, vegetable oil
- A small piece of cork, plastic bead, grape and coin to test
- A kitchen scale, a measuring cup and a spoon
- Sugar and food coloring for a sugar-water layer test (4 cups of water and sugar)
- A notebook and ruler

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Which liquid is used (honey, dish soap, water, oil) and sugar concentration in the sugar-water test
Dependent (what you measure)	The order the liquids stack and the mass of 1/2 cup (120 mL) of each
Controlled (keep the same)	Cup size, temperature, pouring speed, amount of each liquid

Procedure:

1. Put on goggles. Weigh an empty cup, fill it to 1/2 cup (120 mL) with each liquid, weigh again and subtract. This is the mass.
2. Calculate density: mass divided by volume. Write it down.
3. Predict the order from heaviest to lightest.
4. Pour slowly down the inside of the glass, starting with the densest liquid.
5. Drop in small solid objects and note where each stops.
6. Compare the order you saw with your calculation.

Data table to fill in:

LIQUID	MASS OF 1/2 CUP	DENSITY (G/ML)	LAYER (1 = BOTTOM)
Honey			
Dish soap			
Water			
Vegetable oil			

What to graph: A bar graph of density by liquid, and a labeled drawing of the finished column.

Expected result in plain words: Honey sinks to the bottom, then dish soap, then water, then oil on top. A grape or coin settles where its density matches the liquid around it.

> **Safety note**
> Do not drink any liquid used. Wipe spills quickly because honey and oil make floors slippery. Use only the liquids listed.

Extension: Make a sugar-water density rainbow by dissolving 1, 2, 3 and 4 tbsp of sugar in four colored cups of water and layering them.

Project 8: Vitamin C in Juices

Question: Which fruit juice contains the most vitamin C?

Hypothesis example: If a juice is made from citrus fruit, then it will contain the most vitamin C, because oranges and lemons are known to be rich in it.

Materials (all household and safe):

- 1 tsp (5 g) cornstarch
- 1 cup (240 mL) water
- A few drops of tincture of iodine 2% from a pharmacy (adult handles it)
- 1/2 cup (120 mL) each of three juices, such as orange, apple and grape
- A dropper, 4 clear cups, a spoon
- A saucepan and stove for the starch solution (adult only)
- A notebook, labels and a marker

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	The type of juice
Dependent (what you measure)	The number of drops of iodine solution needed before the starch turns dark blue
Controlled (keep the same)	Juice amount, iodine solution strength, drops size, cup size, temperature

Procedure:

1. Put on goggles. An adult boils the cornstarch in 1 cup (240 mL) of water for 1 minute, then lets it cool. Add 4 drops of iodine and 1 cup of water to make a pale blue test solution.
2. Pour 2 tbsp (30 mL) of the test solution into each clean cup.
3. Add 1 tsp (5 mL) of the first juice. The blue color fades as vitamin C reacts.
4. Keep adding the juice, a drop at a time, until the blue just disappears. Count the drops.
5. Repeat for every juice and write down the count.
6. Do 3 trials. Fewer drops of juice means MORE vitamin C.

Data table to fill in:

JUICE	TRIAL 1 DROPS	TRIAL 2 DROPS	TRIAL 3 DROPS	AVERAGE
Orange				
Apple				
Grape				

What to graph: A bar graph of average drops by juice. Label it clearly: lower bar means more vitamin C.

Expected result in plain words: Orange juice usually needs the fewest drops, meaning it has the most vitamin C. Some juices have vitamin C added, so read the label and mention it in your notes.

> **Safety note**
> Iodine stains skin and clothing and is harmful if swallowed. An adult prepares and handles it. Do not drink any juice used in the test, and clean up with gloves.

Extension: Compare fresh juice with juice that has been left open for 3 days to see whether vitamin C breaks down.

Project 9: Natural Indicators Compared

Question: Which natural plant juice is the best acid-base indicator?

Hypothesis example: If a plant juice has strong pigments, then it will show the clearest color change, because pigments change shape in acid or base.

Materials (all household and safe):

- 1/4 cup each of chopped red cabbage, beets and blueberries (or grape skins)
- 1 tsp turmeric powder
- Hot water, handled by an adult
- 3 tbsp (45 mL) each of: white vinegar, tap water, baking soda solution (1 tsp in 1/2 cup)
- 12 small clear cups, strainers and spoons
- Labels and a white background

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Type of plant juice (cabbage, beet, blueberry, turmeric)
Dependent (what you measure)	Number of distinct colors seen across vinegar, water and baking soda
Controlled (keep the same)	Amount of juice, amount of test liquid, light, cup size

Procedure:

1. Put on goggles. An adult makes each juice by soaking the plant in 1/2 cup (120 mL) of hot water for 15 minutes. Strain and cool.
2. Line up 12 cups: 3 per juice, labeled acid, neutral and base.

3. Add 2 tbsp (30 mL) of each juice to its three cups.
4. Add 1 tbsp (15 mL) of vinegar, water and baking soda solution to the matching cups.
5. Photograph the colors under the same light and background.
6. Score each juice by how many clearly different colors it shows.

Data table to fill in:

JUICE	COLOR WITH ACID	COLOR WITH WATER	COLOR WITH BASE	SCORE
Red cabbage				
Beet				
Blueberry				
Turmeric				

What to graph: A bar graph of the score for each indicator, and a photo strip of the colors.

Expected result in plain words: Red cabbage usually wins because it shows pink, purple and green. Turmeric changes from yellow to red-orange only in base. Beet stays mostly red.

> **Safety note**
> Do not taste any of the samples. Turmeric and beet stain. Cover surfaces and wear old clothes.

Extension: Test a mystery liquid with your best indicator and predict whether it is acidic or basic before you see the color.

Project 10: Plastic from Milk

Question: How does the temperature of milk change how much plastic-like solid (casein) forms?

Hypothesis example: If the milk is hotter, then more solid will form, because heat helps the proteins clump when acid is added.

Materials (all household and safe):

- 1 cup (240 mL) whole milk per trial
- 4 tsp (20 mL) white vinegar per trial
- A coffee filter or fine strainer
- A thermometer, a microwave or stove used by an adult
- 3 clear cups, paper towels and a kitchen scale
- Labels and a marker

VARIABLE TYPE	IN THIS PROJECT
Independent (what you change)	Starting temperature of the milk (100 F / 38 C, 140 F / 60 C, 170 F / 77 C)
Dependent (what you measure)	Mass of the solid collected after draining and drying, in grams

VARIABLE TYPE	IN THIS PROJECT
Controlled (keep the same)	Amount of milk, amount of vinegar, stirring time, draining time, type of milk

Procedure:

1. Put on goggles. An adult heats the milk to the chosen temperature and checks it with the thermometer.
2. Pour 1 cup (240 mL) into a cup and add 4 tsp (20 mL) of vinegar. Stir for 1 minute.
3. Let it sit for 3 minutes. Lumps will separate from the liquid.
4. Strain through the coffee filter. Press out the liquid with paper towels.
5. Weigh the solid right away, then let it dry for 24 hours and weigh again.
6. Repeat for each temperature. Do 3 trials.

Data table to fill in:

TEMPERATURE	WET MASS (G)	DRY MASS (G)	NOTES
100 F / 38 C			
140 F / 60 C			
170 F / 77 C			

What to graph: A bar graph of dry mass for each temperature, and photos of the finished solids.

Expected result in plain words: Warm milk usually makes more solid and firmer lumps. The solid is made of the milk protein casein, which clumps when the acid is added. You can press it into a shape and let it harden.

> **Safety note**
> Hot milk can scald. An adult handles all heating and pouring. This is not food: do not eat it, and discard it after the project.

Extension: Press a sample into a cookie cutter and compare hardness after 2 days.

Data and Graphs

Data is the heart of your project. A judge can look at a table and a graph and understand in seconds whether your test was careful.

How to make a good data table

- Give the table a clear title, such as 'Foam height for three catalysts'.
- Put what you changed (independent variable) in the first column.
- Put what you measured (dependent variable) in the other columns, with units in the heading.
- Leave a column for each trial and one for the average.
- Write numbers as you measure them. Never write them from memory later.

Averaging trials

Repeat every setting at least three times. Add the three numbers and divide by three. For example, if foam heights are 6, 7 and 8 inches, the average is (6 + 7 + 8) / 3 = 7 inches. Averages smooth out small errors and show judges that your result is not luck.

Bar graph or line graph?

USE A...	WHEN...	EXAMPLE
Bar graph	You compare separate groups or categories	Foam height for yeast, potato and carrot
Line graph	Your independent variable is a number that changes steadily (time, temperature)	Crystal length over 7 days
Pie chart	Rarely; only for parts of a whole	Percent of each material in a mixture

How to draw a clean graph

1. Put the independent variable on the bottom (x) axis and the dependent variable on the side (y) axis.
2. Label both axes with words and units.
3. Choose even number steps (0, 2, 4, 6...) that fit your largest value.
4. Use a ruler. Color bars or lines differently, with a key.
5. Write a title that says what the graph shows.

Error: what went wrong, honestly

No experiment is perfect. Error is not shameful; it is information. Common sources of error are measuring tools that are not exact, a slightly different amount each time, temperature drift, timing mistakes and tiny differences in the materials. Name two or three sources of error in your conclusion and say how you would reduce them next time. Judges love this.

> **Outliers**
> If one trial is very different from the others, do not erase it. Mark it on your table, write a note about what happened, repeat the trial if possible and explain what you decided to do.

The Research Notebook

The notebook is your proof. It shows the judges that the work is yours and happened over time. It does not have to be pretty. It has to be honest, dated and complete.

What goes in the notebook

- **Title page:** project title, your name, grade and school.
- **Question and hypothesis,** written before you start.

- **Background research:** key facts in your own words, with where you found them.
- **Materials list** and variables table.
- **Procedure,** as numbered steps, with any changes you made and why.
- **Raw data,** with a date for every entry.
- **Photos** of setup and results.
- **Graphs, analysis and conclusion.**
- **Problems and surprises.** These are worth a lot.
- **Reference list.**

Notebook rules

- Write in pen, and cross out mistakes with a single line instead of erasing.
- Date every page.
- Number the pages.
- Write in complete sentences for explanations and in tables for numbers.
- Keep it with you during the experiment.

SAMPLE DAILY ENTRY

☐ Date and time

☐ What I did today

☐ What I measured (numbers and units)

☐ What I noticed (colors, smells, sounds, but never tastes)

☐ What went wrong or surprised me

☐ What I will do next

Board Layout

A standard tri-fold display board is about 36 inches (91 cm) tall and 48 inches (122 cm) wide when open. It has a center panel about 24 inches (61 cm) wide and two side panels about 12 inches (30 cm) wide each. Most fairs allow a board no deeper than 30 inches (76 cm) from front to back, so check the rules.

Keep text large enough to be read from 3 feet (1 m) away: headings at least 1 inch (2.5 cm) tall and body text at least 16 point. Use no more than three colors and plenty of white space. Print or write neatly and glue it straight with a ruler.

PANEL	SIZE	WHAT GOES HERE
Top banner (center)	24 in x 5 in (61 × 13 cm), across the top of the center panel	Title of the project, in the largest letters on the board. Keep it under 10 words.
Left panel	12 in x 30 in (30 × 76 cm)	Question, Hypothesis, Background in 2-3 sentences, Materials list, Variables table

PANEL	SIZE	WHAT GOES HERE
Center panel	24 in x 25 in (61 × 64 cm), under the banner	Procedure (numbered steps with a photo) and the Data tables and Graphs, which are the star of the board
Right panel	12 in x 30 in (30 × 76 cm)	Results, Conclusion, Next steps, References, Acknowledgements
Table in front	Whatever space the fair allows	Notebook, samples, photos, a finished product (for example the crystals or the plastic) and an index card with your name

> **Placement rule**
>
> Eyes read left to right and top to bottom. Put the story in that order: Question and Hypothesis on the left, Procedure and Data in the middle, Results and Conclusion on the right.

Board do's and don'ts

- Do use a neutral background color (white, light gray or pale blue).
- Do use photos with a caption under each.
- Do put the graphs at eye level, between 36 and 60 inches (91-152 cm) from the floor.
- Do not put liquids, chemicals or open containers on the table. Use photos or sealed, labeled samples.
- Do not put your phone number, address or any contact information on the board.
- Do not cover the board with glitter, loose paper or crowded text.

Header Signs With Sample Text

Each section of the board gets a header sign: a plain label in large letters, followed by short text. Below are the ten standard signs, each with sample text using Project 1 (elephant toothpaste catalysts) as the example. Replace the sample with your own words.

HEADER SIGN	SAMPLE TEXT
Title	Which Helper Makes the Biggest Foam? Testing Catalysts in a Hydrogen Peroxide Reaction
Question	Which catalyst makes the tallest foam from 3% hydrogen peroxide: dry yeast, potato or carrot?
Hypothesis	If I use dry yeast, then the foam will be tallest, because yeast has a lot of an enzyme that speeds up the breakdown of hydrogen peroxide.
Materials	3% hydrogen peroxide, dry yeast, raw potato, raw carrot, dish soap, plastic bottle, tray, ruler, timer, goggles
Procedure	1. Pour 1/2 cup of peroxide into a bottle. 2. Add soap. 3. Add the catalyst. 4. Measure the foam after 2 minutes. 5. Repeat 3 times for each catalyst.
Data	Table: foam height in inches for each catalyst, with three trials and the average. Graph: average foam height by catalyst.
Results	Yeast made an average foam height of 9 inches. Potato made 3 inches. Carrot made 2 inches.

HEADER SIGN	SAMPLE TEXT
Conclusion	My hypothesis was supported. Yeast was the best helper. This probably happened because yeast contains a lot of catalase. A source of error was that the bottle was not exactly the same temperature each time. Next, I would test yeast in water at different temperatures.
References	List 2 or 3 books or reliable websites you used, with author, title and year.
Acknowledgements	Thank you to my parent for supervising the experiments and to my teacher for feedback on my graphs.

> **Numbers in the sample**
> The numbers above are an example for the layout only. Always report the values you actually measured.

Presentation Scripts

Speak to the judge, not to the board. Stand beside your board, smile, make eye contact and point to the part you are talking about. It is fine to have a notecard, but try not to read from it.

The 2-minute script

1. **Hello (10 seconds).** 'Hello, my name is ___. My project is called ___.'
2. **Question (15 seconds).** 'I wanted to find out ___ because ___.'
3. **Hypothesis (15 seconds).** 'I predicted that if ___, then ___, because ___.'
4. **Method (30 seconds).** 'I changed ___, measured ___, and kept ___ the same. I repeated each test three times.'
5. **Results (30 seconds).** 'This graph shows ___. The highest average was ___ and the lowest was ___.'
6. **Conclusion (15 seconds).** 'My hypothesis was supported / not supported. I think this happened because ___.'
7. **Close (5 seconds).** 'Thank you. I would be happy to answer any questions.'

The 5-minute script

Use the same structure and add the following extras.

- **Background (30 seconds):** two facts about the science behind your project in your own words.
- **Variables (30 seconds):** name the independent, dependent and controlled variables.
- **Safety (20 seconds):** mention goggles, adult supervision and that nothing was tasted.
- **Demonstration (60 seconds):** show a small sealed sample, a photo sequence or a safe mini-demo if allowed.
- **Error (30 seconds):** name two sources of error and how you would reduce them.
- **Real-world link (20 seconds):** where this science shows up in everyday life.
- **Next steps (20 seconds):** what you would test next.

Practice tips

Practice out loud three times: once alone, once in front of a mirror, and once for a family member who asks you a question at the end. Time yourself. Breathe slowly and pause between sections.

Common Judge Questions and Sample Answers

Judges ask questions to find out how well you understand your own work. A calm, honest answer is better than a long one. If you do not know, say so and describe how you would find out.

JUDGE QUESTION	SAMPLE ANSWER
Why did you choose this project?	I noticed ___ at home and wondered why. I wanted to test it with real measurements.
What was your hypothesis, and was it supported?	I predicted ___. My data supported it because ___. (Or: It was not supported, and I learned ___.)
What was your independent variable?	The one thing I changed on purpose, which was ___.
How did you keep the test fair?	I kept ___, ___ and ___ the same each time, and I changed only ___.
How many trials did you do and why?	Three trials for each setting, so I could average them and see if the result was consistent.
What surprised you?	I did not expect ___. It made me look up ___ to learn more.
What went wrong?	One trial was off because ___. I noted it, repeated it and explain this in my conclusion.
How would you improve this experiment?	I would use a more exact measuring tool, test more settings and repeat more trials.
What is the science behind it?	A short plain-language explanation using two or three key words from your research.
Where is this used in real life?	For example, in ___, ___ or ___.
Did anyone help you?	My parent supervised for safety and my teacher gave feedback. I did the experiment and wrote the report myself.
What would you test next?	I would change ___ to see if ___.

The Judging Rubric

This rubric scores a project out of 40 points across ten criteria. Judges choose a score from 4 (excellent) down to 1 (needs work) for each criterion. Teachers can print it for students to self-check before the fair.

CRITERIA	4 - EXCELLENT	3 - GOOD	2 - DEVELOPING	1 - BEGINNING	POINTS
Question and hypothesis	Clear, testable question; hypothesis with a reason	Clear question; hypothesis with a weak reason	Question is vague or hypothesis has no reason	Missing or untestable	/4
Variables and fair test	All three variable types named and controlled	Variables named, a few not controlled	Only one variable type identified	No fair test	/4
Procedure	Numbered steps anyone could repeat	Steps mostly clear	Steps incomplete	Missing	/4
Data collection	Three or more trials, units, tables filled in honestly	Two trials, tables mostly complete	One trial, few units	Little or no data	/4
Graphs and analysis	Correct graph type, labeled axes, averages explained	Correct graph with small errors	Graph with missing labels	No graph	/4
Conclusion	Supported by data, explains why, names error and next steps	Supported by data, little explanation	Not clearly tied to data	Missing	/4
Scientific understanding	Explains the science in own words with accuracy	Mostly accurate explanation	Partly accurate	Cannot explain	/4
Board and notebook	Neat, organized, readable from 3 feet, all signs present	Mostly neat and organized	Cluttered or missing signs	Hard to follow	/4
Presentation	Confident, clear, within time, answers questions well	Clear, a few pauses	Reads from notes, short answers	Unable to present	/4
Safety and integrity	Safety stated, own work, sources listed	Safety stated, sources mostly listed	Safety unclear, few sources	Unsafe or work not own	/4

Scoring guide: 36-40 points is outstanding; 30-35 is strong; 24-29 is solid; below 24 means the project needs more work before it is ready. Judges should add one comment about what the student did well and one suggestion for next time.

The Final Checklist

BEFORE YOU LEAVE HOME ON FAIR DAY

- ☐ My question is one sentence and testable.
- ☐ My hypothesis uses the if-then-because format.
- ☐ Parent or guardian permission was signed before I started.
- ☐ I used goggles and had an adult present for every experiment.
- ☐ Nothing from my project was tasted or eaten.
- ☐ I changed only one variable and kept the others the same.
- ☐ I repeated every setting at least three times.
- ☐ My data tables have units and averages.
- ☐ My graphs have titles, labeled axes and the right type.
- ☐ My conclusion says whether my hypothesis was supported and why.
- ☐ I named at least two sources of error.
- ☐ All ten header signs are on my board.
- ☐ My board is readable from 3 feet (1 m) away.
- ☐ There are no contact details on my board.
- ☐ My notebook is dated, numbered and complete.
- ☐ I practiced my 2-minute talk out loud.
- ☐ I can answer the common judge questions.
- ☐ I packed tape, glue stick, scissors and extra markers.
- ☐ Any samples are sealed and labeled, with no open liquids.
- ☐ I am ready to smile and enjoy the day.

Next Step

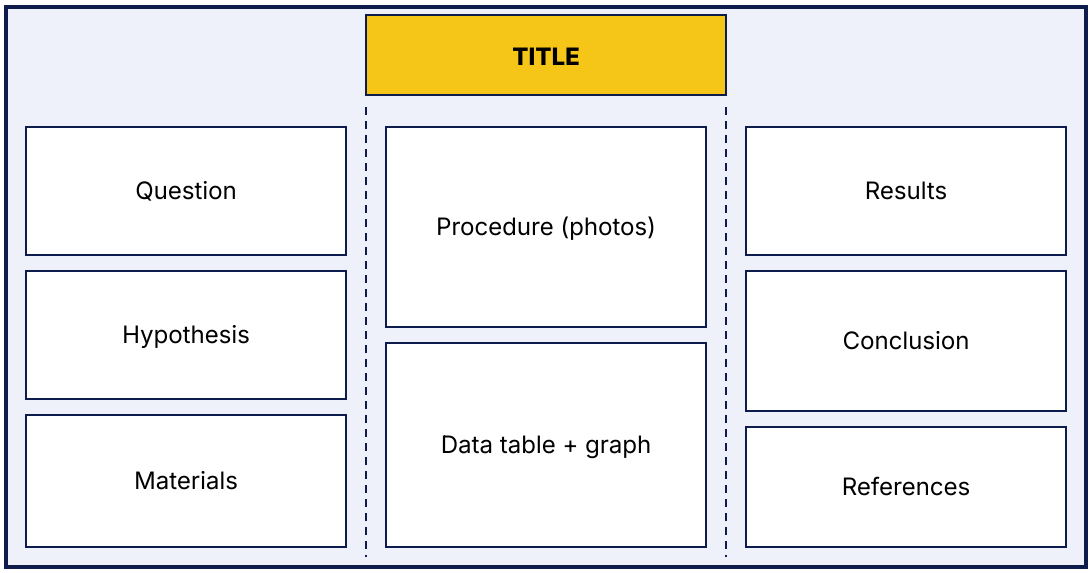
> *Pick one project plan today, write your question in your notebook and mark week 1, day 1 on the calendar.*

Share the plan with the adult who will supervise, get the permission form signed and gather the materials this weekend. Start with the first small step: a clear question, written in one sentence. Everything else in this kit follows from that.

When the fair is over, keep your notebook. A year from now, you can open it, choose one of the extensions and take your project to the next level.

Board layout diagram

A standard tri-fold board (36 in x 48 in when open): left panel for the setup, center panel for the evidence, right panel for the meaning.


TITLE
Question
Hypothesis
Materials
Procedure (photos)
Data table + graph
Results
Conclusion
References

Printable header signs

Print on card stock in landscape, cut out and glue above each section of your board.

TITLE

QUESTION

HYPOTHESIS

MATERIALS

PROCEDURE

DATA

RESULTS

CONCLUSION

REFERENCES

ACKNOWLEDGEMENTS